

ISLAMIC UNIVERSITY OF TECHNOLOGY



COMPUTER NETWORKS LAB
CSE 4512

Lab 3 Report

Submitted by:

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1. Lab Tasks

The objective of this lab is to design and implement a complete network infrastructure for the “**Make Abdullahpur Great Again (MAGA)**” headquarters. The network must support five departments distributed across five floors, ensuring proper IP addressing, DHCP configuration, dynamic routing, and NAT implementation for internet connectivity.

All departments share the following base addressing scheme:

$$192.(i).((i + 3) \bmod 10).0$$

For this lab, the student ID ends with $i = 22$. Therefore, the base network is:

$$192.22.5.0/24$$

2. Device Requirements and Subnet Calculation

Department	Devices	Network / Mask	Subnet Mask	Usable Host Range
Engineering	33	192.22.5.0 /26	255.255.255.192	.1 – .62
Management	30	192.22.5.64 /26	255.255.255.192	.65 – .126
Sales	14	192.22.5.128 /28	255.255.255.240	.129 – .142
Admin	12	192.22.5.144 /28	255.255.255.240	.145 – .158
Marketing	7	192.22.5.160 /28	255.255.255.240	.161 – .174

3. Final Network Topology

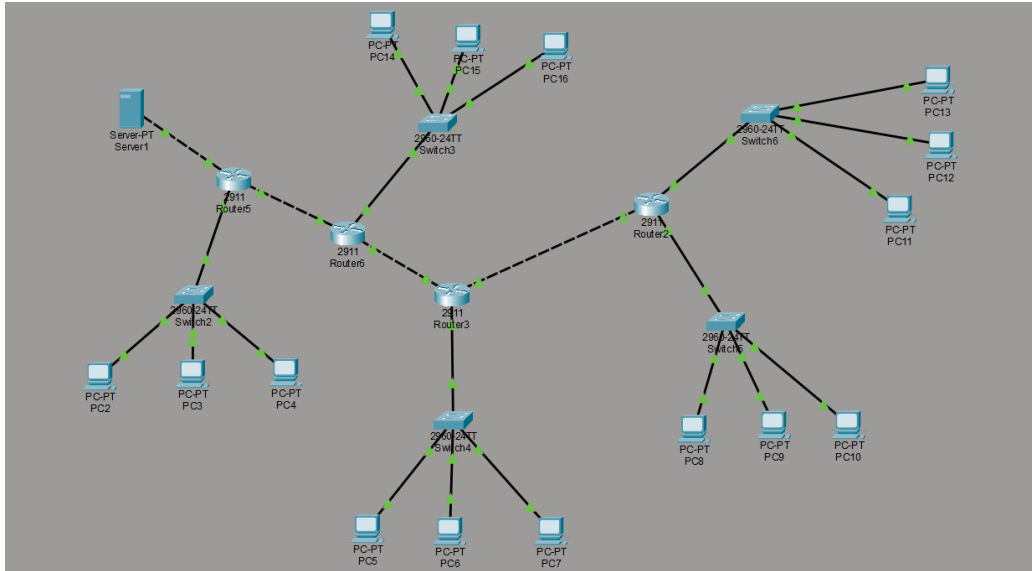


Figure 1: Network Topology Diagram

4. Procedure

1. Device Placement and Initial Setup

All required devices were placed in Cisco Packet Tracer. Three internal routers were used along with an Internet router for NAT configuration. Serial interfaces were used for router-to-router connectivity.

2. Departmental Network Setup

Five switches were added, one for each department. Engineering, Sales and Marketing departments received DHCP services from a dedicated physical DHCP server. Admin and Management departments were configured to receive IP addresses from a router acting as a DHCP server.

3. DHCP Configuration

The DHCP server was assigned a static IP address within the Engineering subnet and configured with separate DHCP pools for Engineering, Sales, and Marketing departments. Router-based DHCP was configured for Admin and Management.

4. Router Interface Configuration

All router interfaces were configured with appropriate IP addresses. RIP version 2 was implemented on all routers to ensure dynamic routing between internal networks.

5. Verification

Connectivity was verified using ping tests between all departments.

5. DHCP Server Configuration

The physical DHCP server was connected to the Engineering switch and manually configured with the following static parameters:

- IP Address: 192.22.5.60
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.22.5.1

Separate DHCP pools were configured to provide IP addresses to Engineering, Sales, and Marketing departments based on their subnet allocations.

6. Router Configuration

Three routers were used to handle internal departmental networks.

7. Observation and Results

DHCP and NAT configuration was flawed. Intra-department communication was confirmed through successful ping tests. Routing tables were dynamically populated using RIP.

8. Challenges Faced

- Careful subnet planning was required to meet different department device requirements.
- Configuring DHCP using both a physical server and router-based services required correct helper address configuration.
- Managing limited router interfaces while connecting all departments required efficient topology planning.